2 a) $R_y(-33,7^\circ) = \begin{pmatrix} \cos(-33,7^\circ) & 0 & \sin(-33,7^\circ) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(-33,7^\circ) & 0 & \cos(-33,7^\circ) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$R_x(-56,3^\circ) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(-56,3^\circ) & \sin(-56,3^\circ) & 0 \\ 0 & -\sin(-56,3^\circ) & \cos(-56,3^\circ) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$T_z(2,36) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2,36 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$R = \begin{pmatrix} 0,83 & 0,46 & -0,31 & 0 \\ 0 & 0,55 & 0,83 & 0 \\ 0,55 & -0,69 & 0,46 & 2,36 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

b) $\vec{p} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad |\vec{p}| = \sqrt{2} \Rightarrow \hat{v} = \begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ 0 \end{pmatrix}$

$D = \cos(30^\circ) \cdot I + (1 - \cos(30^\circ)) \cdot (\vec{v} \cdot \vec{v}^T) - \sin(30^\circ) \cdot \begin{pmatrix} 0 & 0 & \frac{1}{\sqrt{3}} \\ 0 & 0 & -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & 0 \end{pmatrix}$

$= \begin{pmatrix} \cos(30^\circ) & 0 & \frac{1}{\sqrt{3}} \cdot \sin(30^\circ) \\ 0 & \cos(30^\circ) & -\frac{1}{\sqrt{3}} \cdot \sin(30^\circ) \\ -\frac{1}{\sqrt{3}} \cdot \sin(30^\circ) & \frac{1}{\sqrt{3}} \sin(30^\circ) & \cos(30^\circ) \end{pmatrix} + \begin{pmatrix} \frac{1}{2}(1-\cos(30^\circ)) & \frac{1}{2}(1+\cos(30^\circ)) & 0 \\ \frac{1}{2}(1-\cos(30^\circ)) & \frac{1}{2}(1-\cos(30^\circ)) & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$= \begin{pmatrix} \frac{1}{2} + \frac{\sqrt{3}}{4} & \frac{1}{2} - \frac{\sqrt{3}}{4} & \frac{1}{2\sqrt{3}} \\ \frac{1}{2} - \frac{\sqrt{3}}{4} & \frac{1}{2} + \frac{\sqrt{3}}{4} & -\frac{1}{2\sqrt{3}} \\ -\frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} & \frac{\sqrt{3}}{2} \end{pmatrix}$

$e_1^* = \left(\frac{1}{2} - \frac{1}{4\sqrt{3}} \mid \frac{1}{2} + \frac{1}{4\sqrt{3}} \mid \frac{\sqrt{3}}{4}\right)$

$e_2^* = \left(\frac{1}{2} - \frac{1}{4\sqrt{3}} \mid \frac{1}{2} + \frac{1}{4\sqrt{3}} \mid \frac{3\sqrt{3}}{4}\right)$

$e_3^* = \left(\frac{1}{2} - \frac{1}{4\sqrt{3}} \mid \frac{3}{2} - \frac{3}{4\sqrt{3}} \mid \frac{\sqrt{3}}{4}\right)$

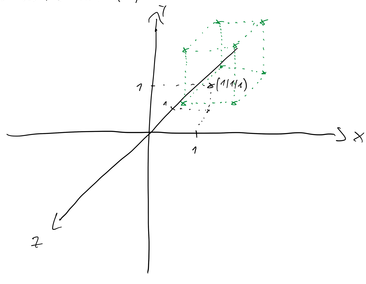
$e_4^* = \left(\frac{3}{2} - \frac{3}{4\sqrt{3}} \mid \frac{1}{2} + \frac{1}{4\sqrt{3}} \mid \frac{\sqrt{3}}{4}\right)$

$e_5^* = \left(\frac{3}{2} - \frac{3}{4\sqrt{3}} \mid \frac{3}{2} - \frac{3}{4\sqrt{3}} \mid \frac{3\sqrt{3}}{4}\right)$

$e_6^* = \left(\frac{3}{2} - \frac{3}{4\sqrt{3}} \mid \frac{3}{2} - \frac{3}{4\sqrt{3}} \mid \frac{\sqrt{3}}{4}\right)$

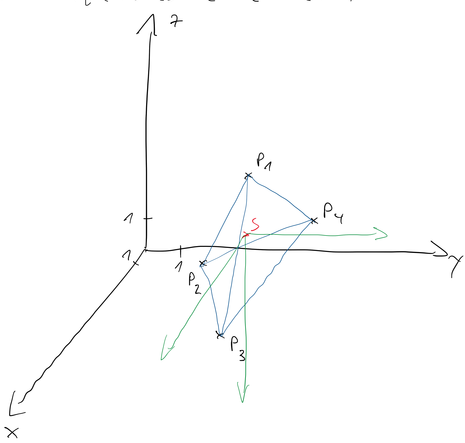
$e_7^* = \left(\frac{3}{2} - \frac{3}{4\sqrt{3}} \mid \frac{1}{2} + \frac{1}{4\sqrt{3}} \mid \frac{3\sqrt{3}}{4}\right)$

$e_8^* = \left(\frac{1}{2} - \frac{3}{4\sqrt{3}} \mid \frac{3}{2} - \frac{3}{4\sqrt{3}} \mid \frac{3\sqrt{3}}{4}\right)$



c) $v_1(5|5|5) \quad v_2(7|5|3) \quad v_3(6|5|1) \quad v_4(6|7|4)$

Schwerpunkt: $\frac{1}{4}\left[\begin{pmatrix} 5 \\ 5 \\ 5 \end{pmatrix} + \begin{pmatrix} 7 \\ 5 \\ 3 \end{pmatrix} + \begin{pmatrix} 6 \\ 5 \\ 1 \end{pmatrix} + \begin{pmatrix} 6 \\ 7 \\ 4 \end{pmatrix}\right] = \begin{pmatrix} 6 \\ 5,5 \\ 3,75 \end{pmatrix}$



Drehung x-Achse: $D_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$

$\Rightarrow v_1' = (-1|-1,75|-1,5)$

$v_2' = (1|0,25|0,5)$

$v_3' = (0|-0,5|-0,5)$

$v_4' = (0|-0,75|1,5)$